

QUESTIONS

Q1

CO3 - AT1 - Analytical Problem Solving
20 marks

Q2

CO3 - AT1 - Analytical Problem Solving
20 marks

Q3

CO3 - AT1 - Analytical Problem Solving
20 marks

Q4

CO3 - AT1 - Analytical Problem Solving
20 marks

Q5

CO3 - AT1 - Analytical Problem Solving
20 marks

5/5 answered

Q1 CO3 - AT1 - Analytical Problem Solving

20 marks

1. Given an $N \times N$ matrix, Write a C Program to determine whether it is:
- a) Symmetric
 - b) Upper triangular
 - c) Lower triangular
 - d) Diagonal matrix
 - e) Identity matrix
 - f) Display all applicable classifications.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, a[20][20];
5     int i, j;
6     int isSymmetric = 1, isUpper = 1, isLower = 1, isDiagonal = 1, isIdentity = 1;
7
8     printf("Enter the size of the matrix (N): ");
9     scanf("%d", &n);
10
11     printf("Enter the elements of the %dx%d matrix:\n", n, n);
12     for (i = 0; i < n; i++) {
13         for (j = 0; j < n; j++) {
14             scanf("%d", &a[i][j]);
15         }
16     }
17
18     // Check Symmetric: a[i][j] == a[j][i]
19     for (i = 0; i < n; i++) {
20         for (j = 0; j < n; j++) {
21             if (a[i][j] != a[j][i]) {
```

Input

5

Output

QUESTIONS

Q1

CO3 - AT2 - Class Test

20 marks



Q2

CO3 - AT2 - Class Test

20 marks



Q3

CO3 - AT2 - Class Test

20 marks



Q4

CO3 - AT2 - Class Test

20 marks



Q5

CO3 - AT2 - Class Test

20 marks



5/5 answered

Q1 CO3 - AT2 - Class Test

20 marks

1. Write a program to cyclically rotate an array by one position to the right.

Sample Input: 1 2 3 4 5

Sample Output: 5 1 2 3 4

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, last;
5
6     scanf("%d", &n);
7
8     int a[n];
9
10    for (i = 0; i < n; i++) {
11        scanf("%d", &a[i]);
12    }
13
14    // Store the last element
15    last = a[n - 1];
16
17    // Shift elements one position to the right
18    for (i = n - 1; i > 0; i--) {
19        a[i] = a[i - 1];
20    }
21
22    // Put last element at the first position
23    a[0] = last;
24
25    // Print rotated array
```

Input

```
5
1 2 3 4 5
```



Output



Visible Test Cases

Test Case 1

